



SIMATS
ENGINEERING



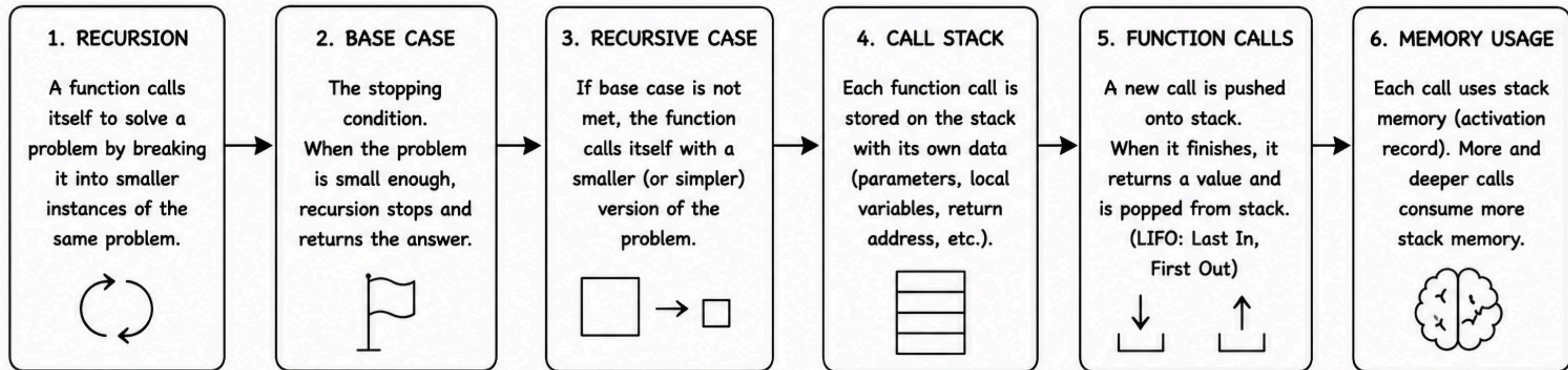
SIMATS
Savoatha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Recursion Concept Map

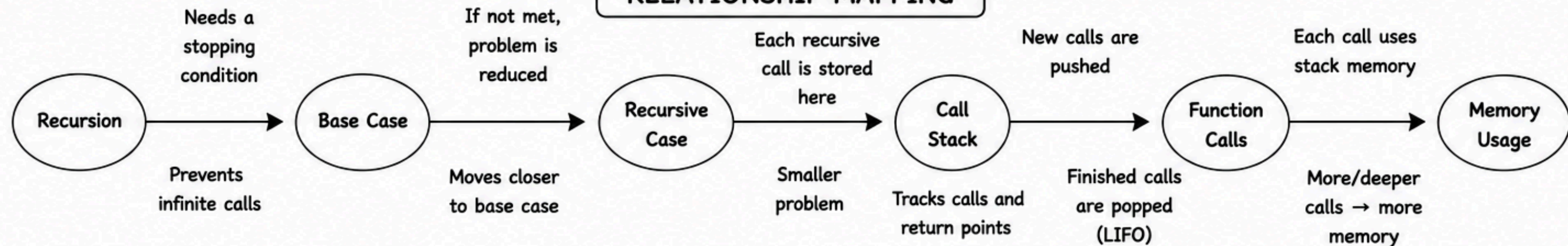
Guided:
DR. Alphonsa

by:
Jeslet Jessica
192524135

RECURSION: HOW IT WORKS



RELATIONSHIP MAPPING



WHY RECURSION IS SUITABLE FOR HIERARCHICAL PROBLEMS

- Hierarchical data is naturally structured in levels (parent → children).
- Recursion matches this structure: solve current part, then recurse on subparts.
- Same logic applies at every level.
- Code becomes simple and easy to understand.
- Ideal for file systems, trees, graphs, nested lists, divide & conquer problems.

KEY TAKEAWAY

Recursion breaks a big problem into smaller versions and uses the call stack to remember where to return. It is perfect for solving hierarchical problems efficiently.

BREAK DOWN → CALL → STACK → RETURN → ANSWER